The Systems Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: GRAPH, SUBSTITUTE, ELIMINATE and MODEL (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (SUBSTITUTE)
- ☐ **A2** (ELIMINATE)
- ☐ **A3** (GRAPH)
- ☐ **A4** (GRAPH)
- ☐ **A5** (MODEL)
- ☐ **A6** (ELIMINATE)

Entrées

10 pts each · choose 4

- ☐ **E1** (SUBSTITUTE)
- ☐ **E2** (ELIMINATE)
- ☐ **E3** (GRAPH)
- ☐ **E4** (MODEL)
- ☐ **E5** (SUBSTITUTE)
- ☐ **E6** (ELIMINATE)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (MODEL)
- ☐ **S2** (GRAPH)
- ☐ **S3** (SUBSTITUTE)
- ☐ **S4** (ELIMINATE)

My skill check (circle each skill you have covered): GRAPH · SUBSTITUTE · ELIMINATE · MODEL

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 SUBSTITUTE ☐ My pick · 5 pts

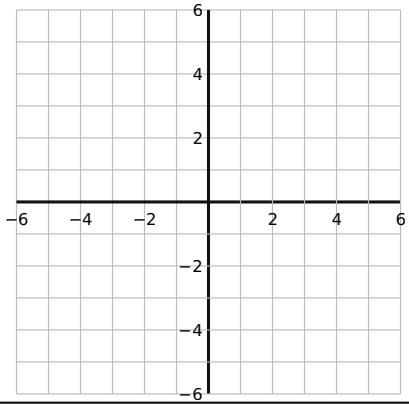
Solve by substitution: $y = 2x - 1$ and $3x + y = 14$

A2 ELIMINATE ☐ My pick · 5 pts

Solve by elimination: $x + y = 11$ and $x - y = 3$

A3 GRAPH ☐ My pick · 5 pts

Graph $y = x + 1$ and $y = -x + 5$ on the grid. Write the solution of the system.



A4 GRAPH ☐ My pick · 5 pts

The system $y = 2x + 3$ and $y = 2x - 4$ has how many solutions? Use the slopes and y-intercepts to explain.

A5 MODEL ☐ My pick · 5 pts

Two numbers have a sum of 20 and a difference of 6. Write a system of equations and find the two numbers.

A6 ELIMINATE ☐ My pick · 5 pts

Solve by elimination: $2x + y = 7$ and $3x - y = 8$

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 SUBSTITUTE ☐ My pick · 10 pts

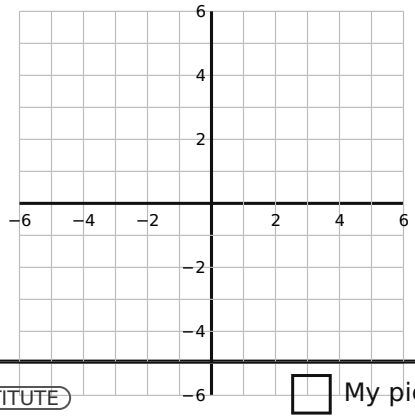
Solve by substitution: $x + 2y = 7$ and $3x - 4y = -9$. Check your answer in **both** equations.

E2 ELIMINATE ☐ My pick · 10 pts

Solve by elimination: $2x + 3y = -1$ and $3x + 5y = -2$

E3 GRAPH ☐ My pick · 10 pts

Graph $2x + y = 4$ and $x - y = -1$ on the grid. Write the solution and check it in both equations.



E4 MODEL ☐ My pick · 10 pts

A school sold 120 tickets to a play. Student tickets cost \$5 and adult tickets cost \$8. The total was \$735. Define variables, write a system, and find how many of each ticket were sold.

E5 SUBSTITUTE ☐ My pick · 10 pts

Dev solved $y = 3x - 2$ and $4x + 2y = 16$ like this:

$4x + 2(3x) - 2 = 16$

$10x - 2 = 16$

$x = \frac{9}{5}$

Find his mistake, then solve the system correctly.

E6 ELIMINATE ☐ My pick · 10 pts

Use elimination to decide how many solutions each system has.

(a) $4x - 6y = 10$ and $-2x + 3y = -5$

(b) $6x + 3y = 12$ and $2x + y = 9$

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

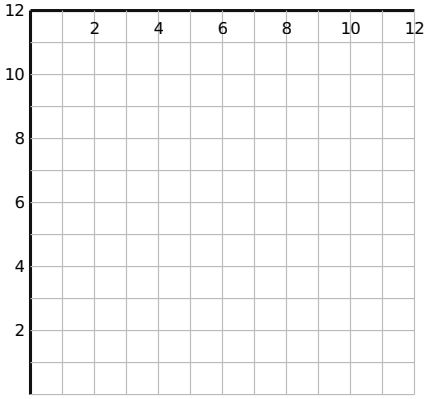
Name: _____

S1 MODEL ☐ My pick · 20 pts

A boat travels 12 miles downstream in 2 hours. The return trip, 12 miles upstream, takes 3 hours. Let b be the boat’s speed in still water and c the current’s speed (miles per hour). **(a)** Write a system. **(b)** Find b and c. **(c)** How long would a 10-mile trip upstream take?

S2 GRAPH ☐ My pick · 20 pts

Plant A is 2 cm tall and grows 2 cm per week. Plant B is 6 cm tall and grows 1 cm per week. Let x = weeks and y = height (cm). **(a)** Write an equation for each. **(b)** Graph both for weeks 0 to 6. **(c)** When are they the same height, and how tall? **(d)** After 8 weeks, which is taller and by how much?



S3 SUBSTITUTE ☐ My pick · 20 pts

Consider $3x + 2y = 8$ and $6x + 4y = c$.

(a) Solve the first equation for y and substitute into the second. Simplify.

(b) For what value of c are there infinitely many solutions? For which c are there none?

(c) When c has that value, give two solutions.

S4 ELIMINATE ☐ My pick · 20 pts

The solution of $ax + 2y = 11$ and $3x + by = 7$ is (3, 1).

(a) Find a and b.

(b) Write the system and solve it by elimination to confirm (3, 1).

(c) Maya says that changing b to 2 would give a system with no solution. Is she right? Explain using elimination.